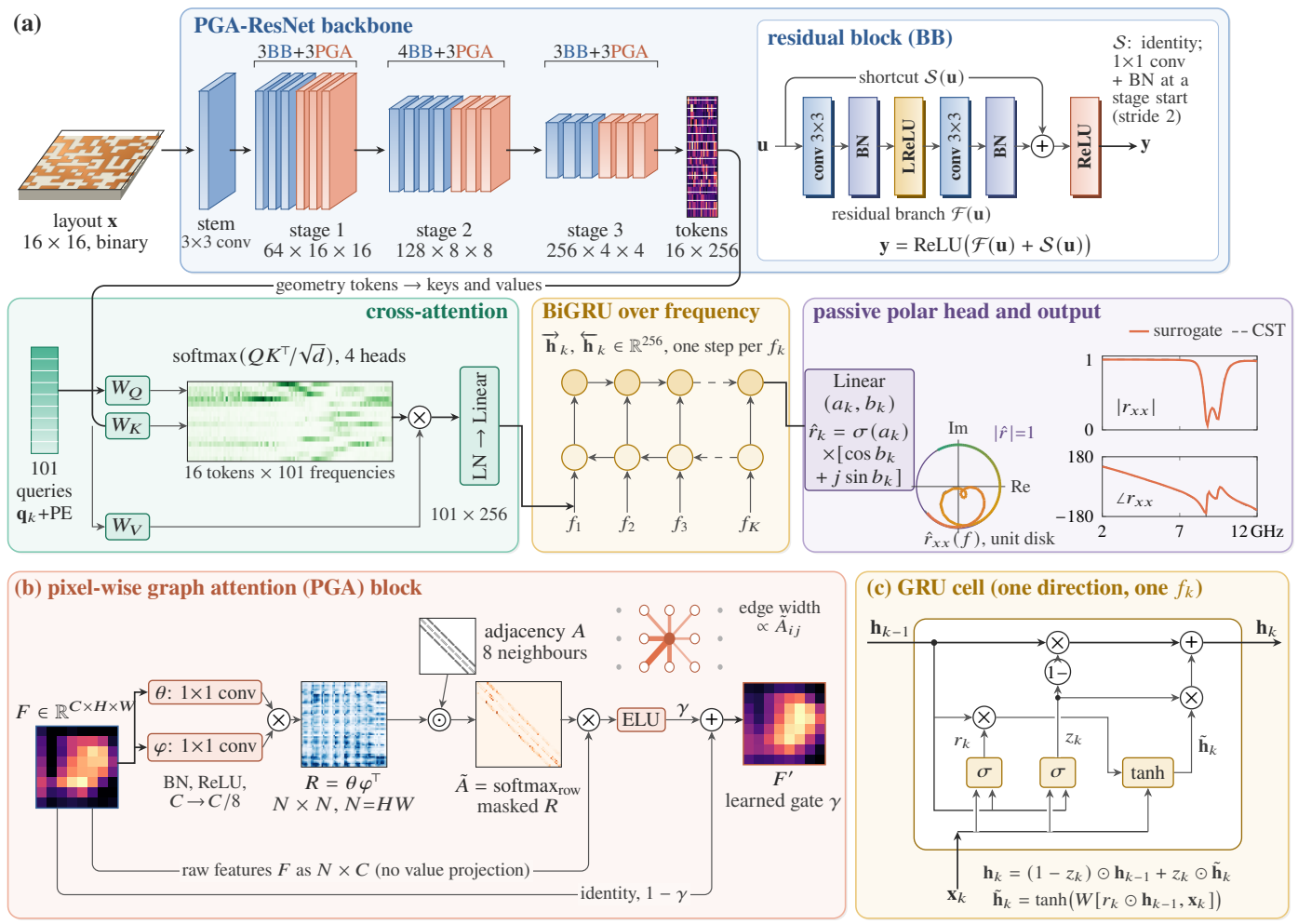

(a)
PGA-ResNet backbone
3BB+3PGA
4BB+3PGA
3BB+3PGA
layout x
16 × 16, binary
stem
3×3 conv
stage 1
64 × 16 × 16
stage 2
128 × 8 × 8
stage 3
256 × 4 × 4
tokens
16 × 256
residual block (BB)
shortcut S(u)
u
conv 3×3
BN
LReLU
conv 3×3
BN
ReLU
y
residual branch F(u)
y = ReLU(F(u) + S(u))
S: identity;
1×1 conv
+ BN at a
stage start
(stride 2)
geometry tokens → keys and values
cross-attention
softmax(QK^T/√d), 4 heads
W_Q
W_K
W_V
16 tokens × 101 frequencies
LN → Linear
101 × 256
101
queries
q_k+PE
BiGRU over frequency
h→_k, h←_k ∈ ℝ^256, one step per f_k
f_1
f_2
f_3
f_K
passive polar head and output
surrogate
CST
Linear
(a_k, b_k)
r̂_k = σ(a_k)
×[cos b_k
+ j sin b_k]
Im
|r̂|=1
Re
r̂_xx(f), unit disk
1
0
|r_xx|
180
−180
∠r_xx
2
7
12 GHz
(b) pixel-wise graph attention (PGA) block
adjacency A
8 neighbours
edge width
∝ Ã_ij
F ∈ ℝ^{C×H×W}
θ: 1×1 conv
φ: 1×1 conv
BN, ReLU,
C → C/8
R = θφ^T
N × N, N=HW
Ã = softmax_row
masked R
ELU
γ
F′
learned gate γ
raw features F as N × C (no value projection)
identity, 1 − γ
(c) GRU cell (one direction, one f_k)
h_{k−1}
h_k
1−
r_k
z_k
h̃_k
σ
σ
tanh
x_k
h_k = (1 − z_k) ⊙ h_{k−1} + z_k ⊙ h̃_k
h̃_k = tanh(W[r_k ⊙ h_{k−1}, x_k])